In Figure 1.4, LO and RO are the lines drawn from the positions of the left and right eyes to the object. These two lines make an angle θ at O. This angle θ is called the angle of parallax.

OL and OR are considered as the radii (x) of a circle. For astronomical calculation, the distance LR = b (basis) can be treated as an arc of this circle, then

$OL = OR = x$

as $LR = b$

$$\theta = \frac{b}{x}$$

Knowing b and θ, x can be calculated which is approximately the distance of the object from the observer.

If the object is the Moon or any near by star, then the angle θ will be too small due to the large astronomical distance and the place of observation. In this case, the two points of observation should be sufficiently spaced on the surface of the Earth.

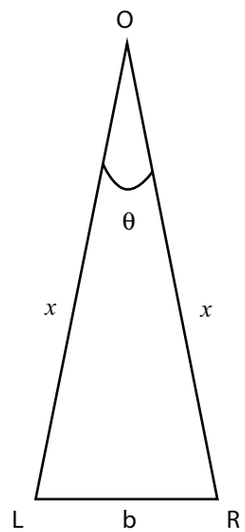


Figure 1.4 Parallax method

Determination of distance of Moon from Earth

In Figure 1.5, C is the centre of the Earth. A and B are two diametrically opposite places on the surface of the Earth. From A and B, the parallaxes θ_1 and θ_2 respectively of Moon M with respect to some distant star are determined with the help of an astronomical telescope. Thus, the total parallax of the Moon subtended on Earth $\angle AMB = \theta_1 + \theta_2 = \theta$.

If θ is measured in radians, then $\theta = \frac{AB}{AM}$; AM≈MC (AM is approximately equal to MC)

$\theta = \frac{AB}{MC}$ or $MC = \frac{AB}{\theta}$; Knowing the values of AB and θ, we can calculate the distance MC of Moon from the Earth.

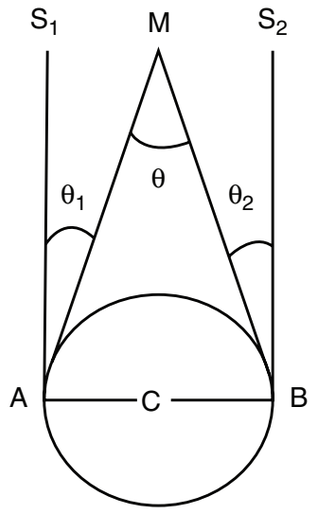


Figure 1.5 Parallax method: determination of distance of Moon from Earth

EXAMPLE 1.2

The Moon subtends an angle of 1° 55′ at the base line equal to the diameter of the Earth. What is the distance of the Moon from the Earth? (Radius of the Earth is $6.4 \times 10^6 m$)

UNIT-1(XI-Physics_Vol-1).indd 14 28-01-2020 13:36:57